LOGICAL STRUCTURES
Lecture 2. Semantics of propositional logic

Advanced Program in Computer Science

Spring, 2025

Contents

3. Propositional logic as a formal language

- Previous section we learned about propositional atoms and how they can be used to build more complex logical formulas.

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  The $\rightarrow e$ rule

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2	p	premise
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still valid if we substitute p with $p \vee \neg r$ and q with $r \rightarrow p$

1	$p \vee \neg r \longrightarrow (r \longrightarrow p)$	premise
2	$p \vee \neg r$	premise
3	$r \longrightarrow p$	$\longrightarrow$e $1, 2$

- Formulas should be strings over the alphabet

$$\{p, q, r, \dots\} \cup \{p_1, p_2, \dots\} \cup \{\neg, \wedge, \vee, \rightarrow, (,)\}$$

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For every well-formed propositional logic formula, the number of left brackets is equal to the number of right brackets.

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Verify that the following formula is well-formed.

$$(((\neg p) \land q) \to (p \land (q \lor (\neg r))))$$

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inversion principle: invert the process of building formulas:

'there is always a unique clause which was used last.'

Inversion Principle

$(((\neg p) \wedge q) \to (p \wedge (q \vee (\neg r))))$

Inversion Principle

$$(((\neg p) \land q) \to (p \land (q \lor (\neg r))))$$

Step 1:

$$(((\neg p) \land q) {\color{blue}\to} (p \land (q \lor (\neg r))))$$

Inversion Principle

$$(((\neg p) \wedge q) \to (p \wedge (q \vee (\neg r))))$$

Step 1:

$$(((\neg p) \wedge q){\color{blue}\to}(p \wedge (q \vee (\neg r))))$$

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Step 3:

$$((({\color{cyan}\neg} p){\color{red}\wedge}q){\color{blue}\to}(p{\color{red}\wedge}(q{\color{cyan}\vee}(\neg r))))$$

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Step 4:

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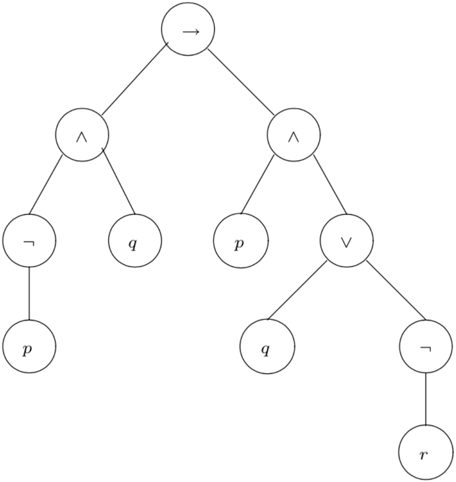

Parse Tree

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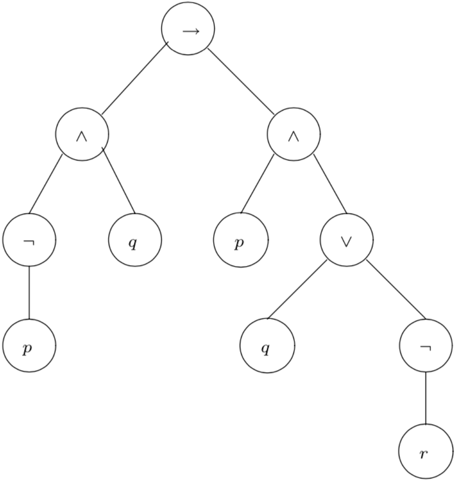
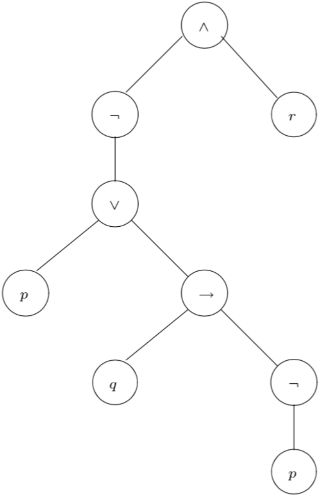
- in case root contains $\neg$: one subtree; others: two subtrees
- the tree constructed inductively.



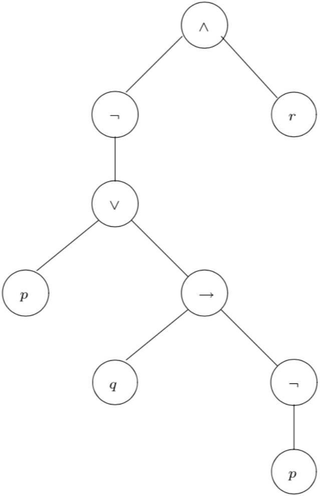
Exercise

What is the formula corresponding to the following parse tree?



Exercise

What is the formula corresponding to the following parse tree?



$$((\neg(p \vee (q \rightarrow (\neg p)))) \wedge r)$$

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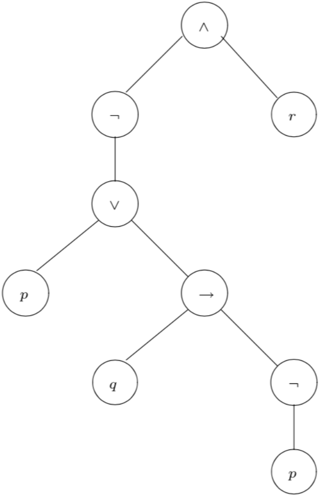

$$\boldsymbol{((\neg(p \vee (q \rightarrow (\neg p)))) \wedge r)}$$

simplify: $\neg(p \vee (q \rightarrow \neg p)) \wedge r$

Definition 3.4

Given the well-formed formula ϕ above, its subformulas are just the ones that correspond to the subtrees of its parse tree.

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$p, q, r, \neg p, \neg p \land q, \neg r, q \lor \neg r, (p \land q) \lor \neg r, \neg p \land q \to (p \land q) \lor \neg r.$

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Find all subformulas of $\neg p \wedge q \to (p \wedge q) \vee \neg r$.

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Given a well-formed formula ϕ, we define its height to be 1 plus the length of the longest path of its parse tree.

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Find the height of $\neg p \wedge q \rightarrow (p \wedge q) \vee \neg r$.

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Exercise

Given the following formulas, draw their corresponding parse tree and find their length.

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1. $p \to \neg q \vee (q \to p)$
2. $\neg((\neg q \wedge (p \to r)) \wedge (r \to q))$
3. $((s \to r \vee t) \vee (\neg q \wedge r) \to \neg(p \to s)) \to r$

Ill-formed formula

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Observe that the string contains $\neg)$ and $\neg$ can not be the rightmost symbol of a well-formed formula.

So $(\neg)() \vee pq \to i$ is not a well-formed formula.

4. Semantics of propositional logic

1. The meaning of logical connectives
2. Soundness of propositional logic
3. Completeness of propositional logic

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Verify the validity of the sequent

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Verify the validity of the sequent

$$p \to q, r \to s \vdash p \lor r \to q \land s$$

- we developed a calculus of reasoning which could verify that sequents of the form $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ are valid.

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- we developed a calculus of reasoning which could verify that sequents of the form $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ are valid.
- verifying the validity of sequents using natural deductions requires a lot of creativity.
- we will now give another account of this relationship between the premises $\phi_1, \phi_2, \ldots, \phi_n$ and conclusion ψ

$$\boxed{\phi_1, \phi_2, \ldots, \phi_n \vDash \psi}$$

$$\boxed{\phi_1, \phi_2, \ldots, \phi_n \;\models\; \psi}$$

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- based on looking at the '***truth values***' of the atomic formulas in the premises and the conclusion

$$\boxed{\phi_1, \phi_2, \ldots, \phi_n \ \models \ \psi}$$

- based on looking at the '***truth values***' of the atomic formulas in the premises and the conclusion
- we combine declarative sentences p and q with a logical connective

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Definition 4.2

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Definition 4.2

1. The set of truth values contains two elements **T** and **F**, where **T** represents 'true' and **F** represents 'false'.
2. A ***valuation*** or ***model*** of a formula ϕ is an assignment of each propositional atom in ϕ to a truth value.

ϕ	ψ	$\phi \wedge \psi$
T	T	T
T	F	F
F	T	F
F	F	F

ϕ	ψ	$\phi \vee \psi$
T	T	T
T	F	T
F	T	T
F	F	F

Truth table for connectives

ϕ	ψ	$\phi \to \psi$
T	T	T
T	F	F
F	T	T
F	F	T

ϕ	$\neg\phi$
T	F
F	T

$\top$
T

$\bot$
F

Evaluation of a logical formula

Example 4.3

Compute the truth table for $(p \to \neg q) \to (q \vee \neg p)$.

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p	q	$\neg p$	$\neg q$	$p \to \neg q$	$q \vee \neg p$	$(p \to \neg q) \to (q \vee \neg p)$
T	T	F	F	F	T	T
T	F	F	T	T	F	F
F	T	T	F	T	T	T
F	F	T	T	T	T	T

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Example 4.5

$\phi \to \psi$ and $\neg\phi \vee \psi$ are semantically equivalent.

4.2. Soundness of propositional logic

We have developed natural deduction calculus and can use natural deduction rules to verify the validity of a sequent.

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Definition 4.6

If, for all valuations in which all $\phi_1, \phi_2, \ldots, \phi_n$ evaluate to **T**, ψ evaluates to **T** as well, we say that

$$\boxed{\boldsymbol{\phi_1, \phi_2, \ldots, \phi_n \vDash \psi}}$$

holds and call $\vDash$ the ***semantic entailment*** relation.

Example 4.7

- $p \wedge q \vdash q$ is valid, then does $p \wedge q \vDash q$ holds?

Example 4.7

- $p \land q \vdash q$ is valid, then does $p \land q \vDash q$ holds?
- What about $p \lor q \vDash q$?

Example 4.7

- $p \wedge q \vdash q$ is valid, then does $p \wedge q \models q$ holds?
- What about $p \vee q \models q$?
- How about $\neg p, p \vee q \models q$?

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- How about $\neg p, p \vee q \vDash q$?
- $\neg p \vDash q \vee \neg q$?

Theorem 4.8 (Soundness)

Let $\phi_1, \phi_2, \ldots, \phi_n$ and ψ be propositional logic formulas. If $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ is valid, then $\phi_1, \phi_2, \ldots, \phi_n \vDash \psi$ holds.

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- $p \wedge q \vdash q$ is valid, then does $p \wedge q \vDash q$ holds?
- What about $p \vee q \vDash q$?
- How about $\neg p, p \vee q \vDash q$?
- $\neg p \vDash q \vee \neg q$?

Theorem 4.8 (Soundness)

Let $\phi_1, \phi_2, \ldots, \phi_n$ and ψ be propositional logic formulas. If $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ is valid, then $\phi_1, \phi_2, \ldots, \phi_n \vDash \psi$ holds.

Proof.

(Textbook, p.46-49) □

4.3. Completeness of propositional logic

The natural deduction rules of propositional logic are **complete**: whenever $\phi_1, \phi_2, \ldots, \phi_n \vDash \psi$ holds, then there exists a natural deduction proof for the sequent $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$.

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Theorem 4.9 (Soundness and Completeness)

Let $\phi_1, \phi_2, \ldots, \phi_n$ and ψ be propositional logic formulas. Then $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ is valid if and only if $\phi_1, \phi_2, \ldots, \phi_n \vDash \psi$ holds.

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Theorem 4.9 (Soundness and Completeness)

Let $\phi_1, \phi_2, \dots, \phi_n$ *and* ψ *be propositional logic formulas. Then* $\phi_1, \phi_2, \dots, \phi_n \vdash \psi$ *is valid if and only if* $\phi_1, \phi_2, \dots, \phi_n \vDash \psi$ *holds.*

Proof.

(Textbook, p.49-53) $\square$

Tautology

Definition 4.10

A formula of propositional logic ϕ is called a ***tautology*** iff it evaluates to $\mathbf{T}$ under all its valuations, i.e. iff $\models \phi$.

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Theorem 4.12

$\phi_1, \phi_2, \ldots, \phi_n$ *and* ψ *be propositional logic formulas. Then*

$$\phi_1, \phi_2, \ldots, \phi_n \models \psi \Leftrightarrow (\phi_1 \wedge \phi_2 \wedge \cdots \wedge \phi_n) \rightarrow \psi \text{ is a tautology}$$

Exercise 1

Which of the following semantic entailments hold?

a) $p \vee q,\ q \models p \rightarrow q$

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a) $p \vee q,\ q \vDash p \to q$

b) $q \vDash p \to q$

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a) $p \lor q,\ q \models p \to q$

b) $q \models p \to q$

c) $p \models p \to q$

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c) $p \models p \to q$

d) $p \to q \models q$

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b) $q \vDash p \to q$

c) $p \vDash p \to q$

d) $p \to q \vDash q$

e) $p \vee q \vDash p$

Exercise 1

Which of the following semantic entailments hold?

a) $p \vee q,\ q \vDash p \to q$

b) $q \vDash p \to q$

c) $p \vDash p \to q$

d) $p \to q \vDash q$

e) $p \vee q \vDash p$

f) $p \to (q \to r) \vDash (p \to q) \to r$

Exercise 1

Which of the following semantic entailments hold?

a) $p \vee q,\ q \models p \to q$

b) $q \models p \to q$

c) $p \models p \to q$

d) $p \to q \models q$

e) $p \vee q \models p$

f) $p \to (q \to r) \models (p \to q) \to r$

g) $(p \to q) \to r \models p \to (q \to r)$

Exercise 1

Which of the following semantic entailments hold?

a) $p \lor q,\ q \models p \to q$

b) $q \models p \to q$

c) $p \models p \to q$

d) $p \to q \models q$

e) $p \lor q \models p$

f) $p \to (q \to r) \models (p \to q) \to r$

g) $(p \to q) \to r \models p \to (q \to r)$

h) $\models (\neg p \to q) \land (\neg q \lor r) \land \neg r \to p$

Exercise 1

Which of the following semantic entailments hold?

a) $p \lor q,\ q \models p \to q$

b) $q \models p \to q$

c) $p \models p \to q$

d) $p \to q \models q$

e) $p \lor q \models p$

f) $p \to (q \to r) \models (p \to q) \to r$

g) $(p \to q) \to r \models p \to (q \to r)$

h) $\models (\neg p \to q) \land (\neg q \lor r) \land \neg r \to p$

i) $\models p \to (q \to \neg p)$

Exercise 2

Check the validity of the following sequents

a) $p \vee q, \;\; \neg p \vdash q$

Exercise 2

Check the validity of the following sequents

a) $p \vee q, \;\; \neg p \vdash q$

b) $p \rightarrow (q \rightarrow r) \vdash q \rightarrow (p \vee r)$

c) $(p \rightarrow q) \rightarrow r \vdash p \rightarrow (q \rightarrow r)$

Exercise 2

Check the validity of the following sequents

a) $p \vee q, \ \neg p \vdash q$

b) $p \to (q \to r) \vdash q \to (p \vee r)$

c) $(p \to q) \to r \vdash p \to (q \to r)$

d) $p \vee (q \wedge r) \vdash p \vee r$

Exercise 2

Check the validity of the following sequents

a) $p \vee q, \ \neg p \vdash q$

b) $p \to (q \to r) \vdash q \to (p \vee r)$

c) $(p \to q) \to r \vdash p \to (q \to r)$

d) $p \vee (q \wedge r) \vdash p \vee r$

e) $a \vee b, \ a \to c, \ \neg d \to \neg b \vdash c \vee d$

Exercise 2

Check the validity of the following sequents

a) $p \vee q, \ \neg p \vdash q$

b) $p \to (q \to r) \vdash q \to (p \vee r)$

c) $(p \to q) \to r \vdash p \to (q \to r)$

d) $p \vee (q \wedge r) \vdash p \vee r$

e) $a \vee b, \ a \to c, \ \neg d \to \neg b \vdash c \vee d$

f) $(a \to b) \wedge (b \to a) \vdash (a \wedge b) \vee (\neg a \wedge \neg b)$

Exercise 2

Check the validity of the following sequents

a) $p \lor q, \ \neg p \vdash q$

b) $p \to (q \to r) \vdash q \to (p \lor r)$

c) $(p \to q) \to r \vdash p \to (q \to r)$

d) $p \lor (q \land r) \vdash p \lor r$

e) $a \lor b, \ a \to c, \ \neg d \to \neg b \vdash c \lor d$

f) $(a \to b) \land (b \to a) \vdash (a \land b) \lor (\neg a \land \neg b)$

g) $\vdash ((p \to q) \to \neg p) \to \neg p$